

GreenLoop — A Neighborhood Tool & Equipment Sharing Cooperative

1. Background

GreenLoop is a community-run cooperative that lets neighbors borrow tools and equipment instead of buying items they'll only use a few times a year — things like drills, ladders, pressure washers, and lawn edgers. Members pay a small annual fee to join, and in return they can borrow any tool in the shared inventory, drop it off at a nearby pickup point, and leave a review afterward. The cooperative wants a database to track members, tools, borrowing activity, drop-off locations, and maintenance history so the organizers can see which tools are popular, which need repair, and which members are the most active borrowers.

2. Business Requirements

1. Anyone who joins becomes a Member, identified by a unique member ID, with a name, email, phone number, and the date they joined.
2. Each Tool has a unique tool ID, name, description, condition, and daily rental fee.
3. Every tool belongs to exactly one Category; one category can contain many tools.
4. Members borrow tools by creating a Loan with checkout date, expected return date, actual return date.
5. A member can have many loans; a tool can appear in many loans over time, but only one active loan at a time.
6. Every loan is picked up from and returned to a Pickup Point with ID, address and hours.
7. One pickup point can serve many loans; each loan uses one pickup point.
8. After returning a tool, a member may optionally leave one Review with rating and comment.
9. A review belongs to exactly one loan; each loan has at most one review.
10. Each Maintenance Record stores date, description and cost.
11. One tool can have many maintenance records; each record belongs to one tool.
12. Maintenance is performed by a Technician with name and contact number.
13. One technician performs many maintenance records; each record has one technician.
14. Member and Technician are separate entities.

3. Additional Business Rules

- Member must exist before a loan references them.
- No loan for tools marked 'needs repair'.
- Tools with loan/maintenance history must be archived instead of deleted.
- Pickup points are independent of members.
- Loan stores the rental fee at the time of borrowing.
- Reviews are optional and only members review tools.

4. ER Diagram Requirements

Include all entities with primary keys, relationships with cardinality and participation, Loan as the associative entity with its attributes, and resolve the many-to-many relationship using Loan.

5. Entity List

Member, Tool, Category, Loan, Pickup Point, Review, Maintenance Record, Technician.

6. Why This Case Is Useful

A compact ER modeling exercise covering keys, relationships, cardinality, participation, and associative entities.